

MATH255: Honours Analysis 2

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Abstract

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1 Metric Spaces

1.1 Openness and closeness

We first give the definition of openness and closeness of sets in metric spaces, and we aim to layout all their equivalent characterizations as fast as possible.

Definition 1 (Open and closed sets). Let (X, d) be a metric space and $A \subseteq X$.

1. We say that A is open in (X, d) if for every $x \in A$, there exists $r > 0$ such that $B(x, r) \subseteq A$.
2. We say that A is closed in (X, d) if $X \setminus A$ is open in (X, d) .

Definition 2 (Closure, interior, and boundary). Let (X, d) be a metric space and $A \subseteq X$.

1. We call the closure of A in (X, d) the set $\overline{A}$ of all points in X such that $\forall r > 0, B(x, r) \cap A \neq \emptyset$.
2. We call the interior of A in (X, d) the set $\text{Int}(A)$ of all points in A such that $\exists r > 0$ such that $B(x, r) \subseteq A$.
3. We call the boundary of A in (X, d) the set $\partial A = \overline{A} \setminus \text{Int}(A)$.

Remark 1. The whole space X and the empty set $\emptyset$ are both open and closed in (X, d) .

Some basic characterizations of the openness and closeness of sets using the definitions of closure and interior are as follows:

Theorem 1 (Open set and interior). *Let (X, d) be a metric space and $A \subseteq X$. Then the following holds:*

1. $\text{Int}(A) \subseteq A$.
2. $\text{Int}(A)$ is open in (X, d) .
3. $B \subset \text{Int}(A)$ for every open set $B \subseteq A$ in (X, d) .
4. Thus, $\text{Int}(A)$ is the largest open set in (X, d) contained in A .

Moreover, A is open in (X, d) iff $\text{Int}(A) = A$.

Theorem 2 (Closed set and closure). *Let (X, d) be a metric space and $A \subseteq X$. Then the following holds:*

1. $A \subseteq \bar{A}$.
2. $\bar{A}$ is closed in (X, d) .
3. $B \supseteq \bar{A}$ for every closed set $B \supseteq A$ in (X, d) .
4. Thus, $\bar{A}$ is the smallest closed set in (X, d) containing A .

Moreover, A is closed in (X, d) iff $A = \bar{A}$.

Convergence with open and close sets

To show further characterizations of the openness and closeness, we need to first define the limit and convergence of sequences in metric spaces.

Definition 3 (Limit and convergence). *Let (X, d) be a metric space, $x \in X$, and $(x_k)_{k \in \mathbb{N}}$ be a sequence of elements in X . We say that $(x_k)_k$ converges to x in (X, d) if and we denote $\lim_{k \rightarrow \infty} x_k = x$ or $x_k \rightarrow x$ as $k \rightarrow \infty$ if*

$$\forall \varepsilon > 0, \exists k_\varepsilon \in \mathbb{N} \text{ such that } \forall k \geq k_\varepsilon, d(x_k, x) < \varepsilon \quad (1)$$

Moreover, x is called the limit of the sequence $(x_k)_k$ in (X, d) .

Definition 4 (Cluster point (limit point)).

Let (X, d) be a metric space, $A \subseteq X$, and $x_0 \in X$.

We say that x_0 is a cluster point of A in (X, d) if $x_0 \in \overline{A \setminus \{x_0\}}$.

Proposition 1. *A point x_0 is a cluster point of a set A iff every open ball $B(x_0, r)$ intersects $A \setminus \{x_0\}$ (namely, contains a point x of A such that $x \neq x_0$). We note that x_0 may not be in A .*

Remark 2. *Let (X, d) be a metric space and $(x_k)_k$ be a sequence in X .*

1. $\lim_{k \rightarrow \infty} x_k = x$ in (X, d) iff $\lim_{k \rightarrow \infty} d(x_k, x) = 0$ as a sequence in $(\mathbb{R}, |\cdot|)$.
2. If $\lim_{k \rightarrow \infty} x_k = x$ in (X, d) , then it's unique
3. Every subsequence of $(x_k)_k$ also converges to the same limit x in (X, d) .

Proposition 2.

x_0 is a cluster point of A iff there exists a sequence $(x_k)_{k \in \mathbb{N}}$ in A such that $\lim_{k \rightarrow \infty} x_k = x_0$ and $x_k \neq x_0$ for all $k \in \mathbb{N}$.

Definition 5 (Isolated point). Let (X, d) be a metric space, $A \subseteq X$, and $x_0 \in A$. We say that x_0 is an isolated point of A in (X, d) if there exists $r > 0$ such that $B(x_0, r) \cap A = \{x_0\}$.

Additional characterizations

We can not show more characterizations of the openness and closeness of sets:

Definition 6 (Derived set). Let (X, d) be a metric space, $A \subseteq X$. The derived set of A in (X, d) is the set A' of all cluster points of A in (X, d) .

Proposition 3. Let (X, d) be a metric space and $A \subseteq X$. Then the closure of A in (X, d) is the union of A and its derived set, i.e. $\overline{A} = A \cup A'$.

Theorem 3 (Close sets in metric spaces). For a metric space (X, d) and $A \subseteq X$, then the set A is closed iff A contain all of its cluster points.

We now characterize the closeness of sets using the convergence of sequences:

Proposition 4. Let (X, d) be a metric space, $x \in X$, and $A \subseteq X$. Then $x \in \overline{A}$ iff there exists a sequence $(x_k)_k$ in A such that $\lim_{k \rightarrow \infty} x_k = x$.

Theorem 4 (Sequential Criterion for Closedness). Let (X, d) be a metric space and $A \subseteq X$. Then the following statements are equivalent:

- A is closed in (X, d) .
- For every $x \in X$ and $(x_k)_{k \in \mathbb{N}}$ sequence of elements in A , if $\lim_{k \rightarrow \infty} x_k = x$ in (X, d) , then $x \in A$.

More properties of open and closed sets includes:

Proposition 5 (Union and intersection of open set). Let (X, d) be a metric space,

1. The intersection of any family $\mathcal{F}$ of open sets in (X, d) is open in (X, d) .
2. The union of any family $\mathcal{F}$ of open sets in (X, d) is open in (X, d) .

Proposition 6 (Union and intersection of closed set). Let (X, d) be a metric space,

1. The intersection of any family $\mathcal{F}$ of closed sets in (X, d) is closed in (X, d) .
2. The union of any finite family $\mathcal{F}$ of closed sets in (X, d) is closed in (X, d) .

Proposition 7 (Set difference of open and closed sets). *Let (X, d) be a metric space,*

1. *If A_1 is an open set in X and A_2 is a closed set, then $A_1 \setminus A_2$ is open in (X, d) .*
2. *If A_1 is a closed set and A_2 is an open set in (X, d) , then $A_1 \setminus A_2$ is closed in (X, d) .*

More characterizations of interior and closure:

Proposition 8. *Let (X, d) be a metric space and $A \subseteq X$*

1. *$\overline{X \setminus A} = X \setminus \text{Int}(A)$, and we have $\text{Int}(X \setminus A) = X \setminus \overline{A}$.*
2. *$\partial A = \overline{A} \setminus \text{Int}(A) = \overline{A} \cap \overline{X \setminus A}$.*

Thus, boundary is the two-way limit point, it's both a limit point of A and a limit point of $X \setminus A$.

1.2 Completeness and compactness

Definition 7 (Cauchy sequence). *Let (X, d) be a metric space and $(x_k)_{k \in \mathbb{N}}$ be a sequence of elements in X . We say that $(x_k)_k$ is a Cauchy sequence in (X, d) if*

$$\forall \varepsilon > 0, \exists k_\varepsilon \in \mathbb{N} \text{ such that } \forall m, n \geq k_\varepsilon, d(x_m, x_n) < \varepsilon \quad (2)$$

Proposition 9. *Every converging sequence in (X, d) is a Cauchy sequence in (X, d) .*

Definition 8 (Completeness, Banach and Hilbert space). *We say that a metric space (X, d) is complete if every Cauchy sequence in (X, d) converges to some limit in (X, d) .*

- *A complete normed vector space is called a Banach space.*
- *A complete inner product space is called a Hilbert space.*

Proposition 10. *Let (X, d) be a metric space and $Y \subseteq X$,*

1. *If (Y, d) is complete, then Y is closed in (X, d) .*
2. *If (X, d) is complete and Y is closed in (X, d) , then (Y, d) is complete.*

Definition 9 (Compactness). *Let (X, d) be a metric space and $A \subseteq X$. We say that A is compact (or sequentially compact) in (X, d) if for every sequence $(x_k)_{k \in \mathbb{N}}$ of elements in A , there exists $x \in A$ and a subsequence $(x_{k_j})_{j \in \mathbb{N}}$ such that $\lim_{j \rightarrow \infty} x_{k_j} = x$ in (X, d) .*

If this is true for $A = X$, then we say that (X, d) is compact (or sequentially compact).

Compactness forces convergence and existence of limits.

Proposition 11. Let (X, d) be a compact metric space and $A \subseteq X$. Then A is compact in (X, d) iff A is closed in (X, d) .

Theorem 5 (Heine-Borel Theorem). Let $X, \|\cdot\|$ be a normed vector space of finite dimension and $A \subseteq X$. Then, A is compact in $(X, \|\cdot\|)$ iff A is closed and bounded in $(X, \|\cdot\|)$.

Proposition 12 (Finite subcover of compact sets).

Let (X, d) be a metric space and $A \subseteq X$. If A is compact in (X, d) , then for every collection $\{U_i\}_{i \in I}$ of open subsets of (X, d) such that $A \subseteq \bigcup_{i \in I} U_i$, there exists a finite sub-collection $\{U_{i_j}\}_{1 \leq j \leq N}$ where $N \in \mathbb{N}$ and $i_j \in I$ for all $1 \leq j \leq N$ such that $A \subseteq \bigcup_{j=1}^N U_{i_j}$.

Corollary 1. Let (X, d) be a metric space and $A \subseteq X$. If A is compact in (X, d) , then for every $r > 0$, there exists a finite number $N_r \in \mathbb{N}$ of elements $x_1, \dots, x_{N_r} \in A$ such that $A \subseteq \bigcup_{j=1}^{N_r} B(x_j, r)$.

Corollary 2. Let $(X, \|\cdot\|)$ be a normed vector space of infinite dimension. Then, the closed unit ball $\{x \in X : \|x\| \leq 1\}$ is not compact in $(X, \|\cdot\|)$. (Hence the Heine-Borel theorem does not hold in infinite dimension.)

Proposition 13. Every normed vector space of finite dimension is Banach.

1.3 Mapping between metric spaces

Definition 10 (Limit of function value). Let (X, d_X) and (Y, d_Y) be two metric spaces, $A \subseteq X$, x_0 be a cluster point of A and $y_0 \in Y$.

1. For $f : A \rightarrow Y$, we say that $f(x)$ converges to y_0 as $x \rightarrow x_0$ and we denote $\lim_{x \rightarrow x_0} f(x) = y_0$ if

$$\forall \varepsilon > 0, \exists \delta_\varepsilon > 0 : \forall x \in A, 0 < d_X(x, x_0) < \delta_\varepsilon \Rightarrow d_Y(f(x), y_0) < \varepsilon \quad (3)$$

2. When $(Y, d_Y) = (\mathbb{R}, |\cdot|)$, for $f : A \rightarrow \mathbb{R}$, we say that $f(x)$ tends to $\pm\infty$ as $x \rightarrow x_0$ and we denote $\lim_{x \rightarrow x_0} f(x) = \pm\infty$ if

$$\forall M > 0, \exists \delta_M > 0 : \forall x \in A, 0 < d_X(x, x_0) < \delta_M \Rightarrow f(x) > M \text{ or } f(x) < -M \quad (4)$$

1.4 Continuous functions on compact sets

2 Infinite Series

Definition 11 (Partial sum). Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$ or $\mathbb{C}$. $k_0 \in \mathbb{Z}$ and $(a_k)_{k \geq k_0}$ by a sequence in X . For every $n \geq k_0$, $S_n = \sum_{k=k_0}^n a_k$ is called the partial sum of order n of $(a_k)_k$.

If $\lim_{n \rightarrow \infty} S_n = S$ in $(X, \|\cdot\|)$, for some $S \in X$, then we write

$$\sum_{k=k_0}^{\infty} a_k = S \quad (5)$$

and we say that the series $\sum_{k=k_0}^{\infty} a_k$ and that S is its sum. If $(S_n)_{n \geq k_0}$ does not converge, then we say that the series $\sum_{k=k_0}^{\infty} a_k$ diverges.

Proposition 14. $\forall k'_0 > k_0$, $\sum_{k=k_0}^{\infty} a_k$ converges iff $\sum_{k=k'_0}^{\infty} a_k$ converges. Therefore, we will sometimes assume without loss of generality that $k_0 = 0$ or 1 .

Proof: $\forall n \geq k'_0$, $\sum_{k=k_0}^n a_k = \sum_{k=k_0}^{k'_0-1} a_k + \sum_{k=k'_0}^n a_k$ where the second term is independent of n . Hence, $\sum_{k=k_0}^{\infty} a_k$ converges iff $\sum_{k=k'_0}^{\infty} a_k$ converges. $\square$

Proposition 15. If $\sum_{k=k_0}^{\infty} a_k$ converges, then $a_k \rightarrow 0$ as $k \rightarrow \infty$.

Proof: If $\sum_{k=k_0}^{\infty} a_k$ converges, then $a_n = S_n - S_{n-1} \rightarrow S - S = 0$ as $n \rightarrow \infty$. $\square$

Proposition 16. If $\sum_{k=k_0}^{\infty} a_k$ and $\sum_{k=k_0}^{\infty} b_k$ converge, then $\forall \lambda, \mu \in \mathbb{R}$ or $\mathbb{C}$, $\sum_{k=k_0}^{\infty} (\lambda a_k + \mu b_k) \rightarrow \lambda \sum_{k=k_0}^{\infty} a_k + \mu \sum_{k=k_0}^{\infty} b_k$.

Proof follows from $\sum_{k=k_0}^n (\lambda a_k + \mu b_k) = \lambda \sum_{k=k_0}^n a_k + \mu \sum_{k=k_0}^n b_k$. $\square$

Theorem 6 (Cauchy criterion for convergence). If $(X, \|\cdot\|)$ is Banach, then $\sum_{k=k_0}^{\infty} a_k$ converges iff $\forall \varepsilon > 0$, $\exists n_\varepsilon \in \mathbb{N}$ such that

$$\forall m > n \geq n_\varepsilon, \left\| \sum_{k=n+1}^m a_k \right\| < \varepsilon \quad (7)$$

Proof: If $(X, \|\cdot\|)$ is Banach, then $(S_n)_{n \geq k_0}$, $S_n = \sum_{k=k_0}^n a_k$ converges iff it is Cauchy, i.e. $\forall \varepsilon > 0$, $\exists n_\varepsilon \in \mathbb{N}$ such that

$$\forall m > n \geq n_\varepsilon, \|S_m - S_n\| = \left\| \sum_{k=n+1}^m a_k \right\| < \varepsilon \quad (8)$$

Remark 3 (Examples).

- **Geometric series** $a_k = r^k$ for some $r \in \mathbb{R}$,

$$S_n = \sum_{k=0}^n r^k = \begin{cases} \frac{1-r^{n+1}}{1-r} & \text{if } r \neq 1 \\ n+1 & \text{if } r = 1 \end{cases} \quad (6)$$

It converges iff $|r| < 1$ with $S = \frac{1}{1-r}$ since $\lim_{n \rightarrow \infty} r^{n+1} = 0$ when $|r| < 1$.

- **Series of matrices** For example, $\sum_{k=0}^{\infty} A^k = (I - A)^{-1}$ for all $n \times n$ real value matrices such that $\|A\|_{2,2} = \max_{\|x\|_2=1} \|Ax\|_2 < 1$.
- **p-series** $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ converges iff $\alpha > 1$.

Definition 12 (Absolute and conditional convergence). $\sum_{k=k_0}^{\infty} a_k$ is said to be absolutely convergent if $\sum_{k=k_0}^{\infty} \|a_k\|$ converges in $(\mathbb{R}, \|\cdot\|)$. It is said to be conditionally convergent if $\sum_{k=k_0}^{\infty} a_k$ converges but $\sum_{k=k_0}^{\infty} \|a_k\|$ diverges.

Proposition 17. If $(X, \|\cdot\|)$ is Banach, then for every series $\sum_{k=k_0}^{\infty} a_k$ in $(X, \|\cdot\|)$, if $\sum_{k=k_0}^{\infty} \|a_k\|$ converges absolutely, then $\sum_{k=k_0}^{\infty} a_k$ converges and

$$\left\| \sum_{k=n}^{\infty} a_k \right\| \leq \sum_{k=n}^{\infty} \|a_k\|, \quad \forall n \geq k_0 \quad (9)$$

Proposition 18. Let $\sum_{k=k_0}^{\infty} a_k$ be a series in a Banach space $(X, \|\cdot\|)$. If $\sum_{k=k_0}^{\infty} a_k$ converges absolutely, then for every bijective function $f : \{k \geq k_0\} \rightarrow \{k \geq k_0\}$ (permutation), the series $\sum_{k=k_0}^{\infty} a_{f(k)}$ converges and

$$\sum_{k=k_0}^{\infty} a_{f(k)} = \sum_{k=k_0}^{\infty} a_k \quad (10)$$

We call the series $\sum_{k=k_0}^{\infty} a_{f(k)}$ a rearrangement of $\sum_{k=k_0}^{\infty} a_k$.

Remark 4.

$\tilde{S}_n = \sum_{k=k_0}^n \|a_k\|$ is nondecreasing: $\tilde{S}_{n+1} \geq \tilde{S}_n$ for all $n \in \mathbb{N}$.

Hence, $\sum_{k=k_0}^{\infty} \|a_k\|$ converges iff $(\tilde{S}_n)_{n \geq k_0}$ is bounded. Moreover, $\sum_{k=k_0}^{\infty} \|a_k\| = \sup_{n \geq k_0} \tilde{S}_n$.

Remark 5.

The converse is true, if $\forall \sum a_k$ that converges absolutely, $\sum a_k$ converges, then $(X, \|\cdot\|)$ is Banach.

2.1 Series with nonnegative term

Theorem 7 (Direct Comparison Test). Let $\sum_{k=k_0}^{\infty} a_k$ and $\sum_{k=k_0}^{\infty} b_k$ be two series in $(\mathbb{R}, \|\cdot\|)$ such that $0 \leq a_k \leq b_k$ for all $k \geq k_0$. Then

1. If $\sum_{k=k_0}^{\infty} b_k$ converges, then $\sum_{k=k_0}^{\infty} a_k$ converges and $\sum_{k=k_0}^{\infty} a_k \leq \sum_{k=k_0}^{\infty} b_k$.
2. If $\sum_{k=k_0}^{\infty} a_k$ diverges, then $\sum_{k=k_0}^{\infty} b_k$ diverges.

Corollary 3 (Limit Comparison Test). Let $\sum_{k=k_0}^{\infty} a_k$ and $\sum_{k=k_0}^{\infty} b_k$ be two series in $(\mathbb{R}, \|\cdot\|)$ such that $a_k > 0$ and $b_k > 0$ for all $k \geq k_0$, and $\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \ell \in (0, \infty)$. Then $\sum_{k=k_0}^{\infty} a_k$ converges if and only if $\sum_{k=k_0}^{\infty} b_k$ converges.

Theorem 8 (Root Test). Let $\sum_{k=k_0}^{\infty} a_k$ be a series in $(\mathbb{R}, \|\cdot\|)$ such that $a_k \geq 0$ for all $k \geq k_0$.

1. If $\limsup_{k \rightarrow \infty} \sqrt[k]{a_k} < 1$, then $\sum_{k=k_0}^{\infty} a_k$ converges.
 2. If $\limsup_{k \rightarrow \infty} \sqrt[k]{a_k} > 1$, then $\sum_{k=k_0}^{\infty} a_k$ diverges.
- If $\limsup_{k \rightarrow \infty} \sqrt[k]{a_k} = 1$, then the test is inconclusive.

Proposition 19 (Ratio Test). Let $\sum_{k=k_0}^{\infty} a_k$ be a series in $(\mathbb{R}, \|\cdot\|)$ such that $a_k > 0$ for all $k \geq k_0$.

1. If $\limsup_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1$, then $\sum_{k=k_0}^{\infty} a_k$ converges.
2. If $\liminf_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} > 1$, then $\sum_{k=k_0}^{\infty} a_k$ diverges.

Proof: (1) Assume that $R = \limsup_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1$. Then $\forall \varepsilon \in (0, 1 - R)$, $\exists n_\varepsilon \in \mathbb{N}$ such that $\forall k \geq n_\varepsilon, \sup_{k \geq n} \frac{a_{k+1}}{a_k} < R + \varepsilon < 1$, which gives for all $k \geq n_\varepsilon$, $a_{k+1} < (R + \varepsilon)a_k < (R + \varepsilon)^{k-n_\varepsilon+1}a_{n_\varepsilon}$. When $\varepsilon \in (0, 1 - R)$, $0 < R + \varepsilon < 1$, so $\sum_{k=n_\varepsilon}^{\infty} (R + \varepsilon)^{k-n_\varepsilon+1}a_{n_\varepsilon}$ converges, which by direct comparison implies that $\sum_{k=k_0}^{\infty} a_k$ converges.

(2) Assume that $R = \liminf_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} > 1$. Then $\exists k_1 \geq k_0$ such that $\forall k \geq k_1, \inf_{k \geq n} \frac{a_{k+1}}{a_k} > R - \varepsilon > 1$, i.e. $a_{k+1} > a_k$, which implies $a_k \not\rightarrow 0$ as $k \rightarrow \infty$, thus $\sum_{k=k_0}^{\infty} a_k$ diverges. $\square$

2.2 Power series

Now we focus on series in $(\mathbb{R}, |\cdot|)$ with terms of different sign.

Theorem 9 (Alternating Series Test). Let $\sum_{k=k_0}^{\infty} a_k$ be a series in $(\mathbb{R}, |\cdot|)$ such that $0 \leq a_{k+1} \leq a_k$ for all $k \geq k_0$ and $\lim_{k \rightarrow \infty} a_k = 0$. Then $\sum_{k=k_0}^{\infty} a_k$ converges and

$$0 \leq \sum_{k=k_0}^{\infty} (-1)^{k-k_0} a_k \leq a_{k_0} \quad (11)$$

Example 1

$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}$ converges since $0 \leq \frac{1}{k+1} \leq \frac{1}{k}$ for all $k \geq 1$ and $\lim_{k \rightarrow \infty} \frac{1}{k} = 0$.

Proof: By observing that

$$\sum_{k=k_0}^{\infty} (-1)^k a_k = \sum_{k=0}^{\infty} (-1)^{k+k_0} a_{k+k_0} \quad (12)$$

$$= (-1)^{k_0} \sum_{k=0}^{\infty} (-1)^k a_{k+k_0} \quad (13)$$

we can assume that $k_0 = 0$.

For every $n \in \mathbb{N}$,

$$S_{2n} = a_0 - a_1 + a_2 - \cdots + a_{2n-2} - a_{2n-1} + a_{2n} \quad (14)$$

$$= (a_0 - a_1) + (a_2 - a_3) + \cdots + (a_{2n-2} - a_{2n-1}) + a_{2n} \quad (15)$$

$$\geq 0 \quad (16)$$

$$S_{2n+1} = a_0 - a_1 + a_2 - \cdots + a_{2n-2} - a_{2n-1} + a_{2n} - a_{2n+1} \quad (17)$$

$$= (a_0 - a_1) + (a_2 - a_3) + \cdots + (a_{2n-2} - a_{2n-1}) + (a_{2n} - a_{2n+1}) \quad (18)$$

$$\leq 0 \quad (19)$$

Definition 13 (Cauchy product). Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be series in $(\mathbb{R}, |\cdot|)$ or $(\mathbb{C}, |\cdot|)$.

For each $n \in \mathbb{N}$, define

$$c_n = \sum_{k=0}^n a_k b_{n-k} = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0 \quad (20)$$

Then the series $\sum_{n=0}^{\infty} c_n$ is called the Cauchy product of $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$.

Proposition 20. If $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ are series in $(\mathbb{R}, |\cdot|)$ or $(\mathbb{C}, |\cdot|)$ such that $\sum_{k=0}^{\infty} a_k$ converges and $\sum_{k=0}^{\infty} b_k$ converges (not necessarily absolutely). Then the Cauchy product $\sum_{n=0}^{\infty} c_n$ converges and $\sum_{n=0}^{\infty} c_n = (\sum_{k=0}^{\infty} a_k)(\sum_{k=0}^{\infty} b_k)$.

Example 2 (Counter example of Cauchy product of two conditionally converging series)

Consider $a_k = b_k = \frac{(-1)^k}{\sqrt{k+1}}$. Then $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ converge by the alternating series test and p -series convergence.

In this case:

$$|c_n| = \left| \sum_{k=0}^n a_k b_{n-k} \right| \quad (21)$$

$$(22)$$

Definition 14 (Power series). Let $(a_k)_k$ be a sequence in $\mathbb{R}$ or $\mathbb{C}$ and $x_0 \in \mathbb{R}$ or $\mathbb{C}$. The function $x \mapsto \sum_{k=0}^{\infty} a_k (x - x_0)^k$ is defined at points where it converges, is called a power series.

Proposition 21 (Radius of convergence). Let $\sum_{k=0}^{\infty} a_k (x - x_0)^k$ be a power series in $\mathbb{R}$ or $\mathbb{C}$. Define $R = \frac{1}{\limsup_{k \rightarrow \infty} \sqrt[k]{|a_k|}}$ with the convention that $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$. Then R is called the radius of convergence of the power series, and

$$\text{if } |x - x_0| < R, \text{ then } \sum_{k=0}^{\infty} a_k (x - x_0)^k \text{ converges,} \quad (23)$$

$$\text{if } |x - x_0| > R, \text{ then } \sum_{k=0}^{\infty} a_k (x - x_0)^k \text{ diverges.} \quad (24)$$

Proposition 22. Let $f(x) = \sum_{k=0}^{\infty} a_k(x - x_0)^k$ be a power series in $(\mathbb{R}, |\cdot|)$ with the radius of convergence $R > 0$. Then f is infinitely differentiable in $(x_0 - R, x_0 + R)$

Example 3

For every $x \in (0, 2)$, we can write $\ln x = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} (x-1)^k$.

Indeed, let $\tilde{\ln} x = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} (x-1)^k$. We want to show that $\tilde{\ln} x = \ln x$, i.e., $\tilde{\ln} e^y = y$ where $y = \ln x$. Observe that

$$\tilde{\ln}'(x) = \sum_{k=1}^{\infty} (-1)^{k-1} (x-1)^{k-1} = \sum_{k=1}^{\infty} (1-x)^{k-1} \quad (25)$$

$$= \sum_{k=0}^{\infty} (1-x)^k \quad (26)$$

$$= \frac{1}{1-(1-x)} = \frac{1}{x} \quad (27)$$

More, this is true for all $x \in (0, 2)$ since the radius of convergence of $\sum_{k=0}^{\infty} (1-x)^k$ is 1. It follows that

$$\frac{d}{dy} [\tilde{\ln} e^y - y] = e^y \tilde{\ln}' e^y - 1 = e^y \frac{1}{e^y} - 1 = 0 \quad (28)$$

hence $\tilde{\ln} e^y - y = \tilde{\ln} e^0 - 0 = 0$, i.e. $\tilde{\ln} x = \ln x$. $\square$

Definition 15 (Taylor series). Let I be an interval in $\mathbb{R}$, $x_0 \in I$ and $f : I \rightarrow \mathbb{R}$ be infinitely differentiable at x_0 . Then the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \quad (29)$$

is called the Taylor series of f at x_0 .

Theorem 10 (Taylor's theorem with mean value reminder).

Let $x, x_0 \in \mathbb{R}$, $x \neq x_0$, $I = \begin{cases} [x_0, x] & \text{if } x > x_0 \\ [x, x_0] & \text{if } x < x_0 \end{cases}$, $\text{int}(I) =$

$\begin{cases} (x_0, x) & \text{if } x > x_0 \\ (x, x_0) & \text{if } x < x_0 \end{cases}$ and $f : I \rightarrow \mathbb{R}$ be n times continuously differentiable in I (i.e. $f^{(n)}$ exists and is continuous in I). and $n + 1$ times differentiable in $\text{int}(I)$.

Then, $\exists x' \in \text{int}(I)$ such that

$$f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k}_{\text{partial sum of Taylor series at } x_0} + \underbrace{\frac{f^{(n+1)}(x')}{(n+1)!} (x - x_0)^{n+1}}_{\text{mean value reminder}} \quad (30)$$

We've proven that every power series $f(x) = \sum_{k=0}^{\infty} a_k(x - x_0)^k$ is equal to its Taylor series at $x_0 \in (x_0 - R, x_0 + R)$.

However, this is not true in general where f is only assumed to be infinitely differentiable at x_0 . For example,

Example 4

The function $x \mapsto e^{-\frac{1}{x^2}}$ for $x \neq 0$ and 0 for $x = 0$ is infinitely differentiable in $\mathbb{R}$ and such that $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}$. That means that its Taylor series at 0 is the zero series.

Thus the function itself is not equal to its Taylor series at 0.

Define $g(t) = f(x) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (t-t)^k$. We want to show that $g(x_0) = f(x) - \frac{1}{(n+1)!} f^{(n+1)}(x')(x-x_0)^{n+1}$ for some $x' \in \text{int}(I)$.

In particular, $g(x) = f(x) - f(x) = 0$.

g is continuous on I and differentiable on $\text{int}(I)$ (by assumption on f) and

$$g'(t) = \sum_{k=0}^n \frac{f^{(k+1)}(t)}{k!} (t-t)^k \quad (31)$$

$$= \frac{f^{(n+1)}(t)}{n!} (t-t)^n \quad (32)$$

$$= \frac{f^{(n+1)}(t)}{(n+1)!} (t-x_0)^{n+1} - \frac{f^{(n+1)}(t)}{(n+1)!} (t-x)^{n+1} \quad (33)$$

Now define

$$h(t) = g(t)(x-x_0)^{n+1} - g(x_0)(x-t)^{n+1} \quad (34)$$

so that

$$h(x) = 0 \quad (35)$$

$$h(x_0) = 0 \quad (36)$$

Since h is continuous on I and differentiable on $\text{int}(I)$ (by assumption on g), and

The point here is to construct a function h such that it vanishes at both x and x_0 so that we can apply Rolle's theorem to it.

$$h'(t) = g'(t)(x-x_0)^{n+1} + (n+1)g(x_0)(x-t)^n \quad (37)$$

$$(38)$$

By Rolle's Theorem, it follows that

$$\exists x' \in \text{int}(I) \text{ such that } h'(x') = 0 \quad (39)$$

3 Riemann-Stieltjes integral

Riemann-Stieltjes integral generalizes the Riemann integral by introducing a weighting function α . The standard width in the Riemann integral $\Delta x_i = x_i - x_{i-1}$ is replaced by $\Delta \alpha_i = \alpha(x_i) - \alpha(x_{i-1})$.

This has lots of (physical) implications. For example, α can be a cumulative distribution function of a random variable, and then the Riemann-Stieltjes integral of a function f with respect to α becomes the expectation.

In fact, it is a reformulation of what we usually write as $\int f(x)\rho(x)dx$ in physics or probability.

We first give definition of the Riemann-Stieltjes integral, then we clarify the keep intuition behind it.

Definition 16 (Partition). Let $a, b \in \mathbb{R}$ such that $a < b$. We call partition of $[a, b]$ a finite set $\{x_0, x_1, \dots, x_n\}$, where $n \in \mathbb{N}$, of elements in $[a, b]$ such that $a = x_0 \leq x_1 \leq \dots \leq x_n = b$.

Definition 17 (Riemann-Stieltjes integrals). Let $\alpha, f : [a, b] \rightarrow \mathbb{R}$ be such that α is increasing on $[a, b]$ and f is bounded on $[a, b]$. For every partition $P = (x_0, x_1, \dots, x_n)$ of $[a, b]$, we define

$$L(P, f, \alpha) = \sum_{i=1}^n m_i \Delta \alpha_i, \quad U(P, f, \alpha) = \sum_{i=1}^n M_i \Delta \alpha_i \quad (40)$$

where $\Delta \alpha_i = \alpha(x_i) - \alpha(x_{i-1})$, $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$ and $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$. We then define

$$\int_a^b f d\alpha = \sup_P L(P, f, \alpha), \quad \overline{\int_a^b f d\alpha} = \inf_P U(P, f, \alpha) \quad (41)$$

they are called the **lower and upper Riemann-Stieltjes integrals** of f with respect to α on $[a, b]$ respectively.

If $\int_a^b f d\alpha = \overline{\int_a^b f d\alpha}$, then we denote $\int_a^b f d\alpha = \int_a^b f d\alpha = \overline{\int_a^b f d\alpha}$ and we say that f is **Riemann-Stieltjes integrable** with respect to α on $[a, b]$, and $\int_a^b f d\alpha$ is called the Riemann-Stieltjes integral of f with respect to α on $[a, b]$.

We denote by $\mathcal{R}([a, b], \alpha)$ the set of all bounded functions $f : [a, b] \rightarrow \mathbb{R}$ which are Riemann-Stieltjes integrable with respect to α on $[a, b]$.

In the case when $\alpha(x) = x$, we say Riemann integral instead of Riemann-Stieltjes integral and we write $\int_a^b f = \int_a^b f d\alpha$, $L(P, f) = L(P, f, \alpha)$, $\mathcal{R}([a, b]) = \mathcal{R}([a, b], \alpha)$ and so on.

Remark 6. α is bounded on $[a, b]$ since $\alpha(a) \leq \alpha(x) \leq \alpha(b) \forall x \in [a, b]$ and $\alpha(a), \alpha(b) \in \mathbb{R}$.

We note that we have:

$$\left(\inf_{a \leq x \leq b} f(x) \right) (\alpha(b) - \alpha(a)) = \left(\inf_{a \leq x \leq b} f(x) \right) \sum_{i=1}^m \Delta \alpha_i \quad (42)$$

$$\leq L(P, f, \alpha) \leq U(P, f, \alpha) \quad (43)$$

$$\leq \left(\sup_{a \leq x \leq b} f(x) \right) (\alpha(b) - \alpha(a)) \quad (44)$$

since $\inf_{a \leq x \leq b} f(x) \leq m_i \leq M_i \leq \sup_{a \leq x \leq b} f(x)$ and $\Delta \alpha_i \geq 0$, hence,

$$\left(\inf_{a \leq x \leq b} f(x) \right) (\alpha(b) - \alpha(a)) \leq \int_a^b f d\alpha \leq \int_a^b f d\alpha \leq \left(\sup_{a \leq x \leq b} f(x) \right) (\alpha(b) - \alpha(a)) \quad (45)$$

Unlike the Riemann integral, the Riemann-Stieltjes integral is constructed through sets argument with inf/sup instead of sequences argument with limits. The core of the construction is built on the sets of partitions, which then gives sets of lower and upper sums. Instead of taking limits of $n \rightarrow \infty$ in the Riemann integral, we take the supremum and infimum over the sets of partitions in the Riemann-Stieltjes integral.

Here, we highlight that taking inf/sup works like taking a generalized limit (but on a set). (Definition of Inf/sup and the process of taking them naturally inherits (wraps/encapsulates) the epsilon-delta style of arguments in the sequential limit). Moreover, these set arguments are built on the technical tools of refinement. Refinement is like moving forward toward the direction of limit, (increasing n when $n \rightarrow \infty$). We will see how it works in the proof of the following propositions and the integrability criterion.

Definition 18 (Refinement). *We say that a partition P^* of $[a, b]$ is a refinement of a partition P of $[a, b]$ if $P^* \supseteq P$. Given two partitions P_1 and P_2 of $[a, b]$, we say that $P_1 \cup P_2$ is a common refinement of P_1 and P_2 .*

The inclusion $P^* \supseteq P$ works like the inequality $N \geq N_0$ in the sequential limit. Taking the common refinement of two partitions is like taking the $N = \max(N_1, N_2)$ in the epsilon-delta style of arguments in the sequential limit.

Proposition 23. *If P^* is a refinement of P , then*

$$L(P, f, \alpha) \leq L(P^*, f, \alpha) \text{ and } U(P^*, f, \alpha) \leq U(P, f, \alpha) \quad (46)$$

Proof: We prove the results in the case where P^* contain one more point x^* than P (the general case then follows by induction).

Assume that $x^* \in (x_{i-1}, x_i)$ where (x_{i-1}, x_i) are consecutive points of P . Then

$$L(P^*, f, \alpha) - L(P, f, \alpha) = m_i^* \Delta \alpha_i^* + m_{i+1}^* \Delta \alpha_{i+1}^* - m_i \Delta \alpha_i \quad (47)$$

where $m_i^* = \inf_{x \in [x_{i-1}, x^*]} f(x)$, $m_{i+1}^* = \inf_{x \in [x^*, x_i]} f(x)$, $\Delta\alpha_i^* = \alpha(x^*) - \alpha(x_{i-1})$ and $\Delta\alpha_{i+1}^* = \alpha(x_i) - \alpha(x^*)$.

Since $[x_{i-1}, x^*] \subseteq [x_{i-1}, x_i]$ and $[x^*, x_i] \subseteq [x_{i-1}, x_i]$, we have

$$m_i \leq m_i^*, m_i \leq m_{i+1}^* \quad (48)$$

Moreover, since α is increasing, we have $\Delta\alpha_i^*, \Delta\alpha_{i+1}^* \geq 0$ (so $m_i \Delta\alpha_i^* \leq m_i^* \Delta\alpha_i^*$).

Therefore, we obtain

$$L(P^*, f, \alpha) - L(P, f, \alpha) \geq m_i(\Delta\alpha_i^* + \Delta\alpha_{i+1}^* - \Delta\alpha_i) = 0 \quad (49)$$

Thus, $L(P^*, f, \alpha) \geq L(P, f, \alpha)$, and similarly, we can obtain $U(P^*, f, \alpha) \leq U(P, f, \alpha)$. $\square$

Proposition 24. $\int_a^b f d\alpha \leq \overline{\int_a^b f d\alpha}$

Proof: $\forall P_1, P_2$ partitions of $[a, b]$, let $P^* = P_1 \cup P_2$ be a common refinement of P_1 and P_2 . Then by the previous proposition,

$$L(P_1, f, \alpha) \leq L(P^*, f, \alpha) \leq U(P^*, f, \alpha) \leq U(P_2, f, \alpha) \quad (50)$$

where the middle inequality follows from infimum is less than or equal to supremum.

Hence, $\int_a^b f d\alpha \leq \overline{\int_a^b f d\alpha}$ follows by taking the supremum over P_1 and the infimum over P_2 .

(Here, taking the inf/sup is like taking the limit.) $\square$

Indeed, the refinement gives control on the approximation of the lower and upper sums to the integral. In the above proposition, we show the gap between the lower and upper integrals is always exists by inserting common refinement in the middle. In turn, we can also bound the error in the approximation (gap in refinement) by controlling the outside gap between the lower and upper sums on previous partitions.

The idea here is like taking $N = \max(N_1, N_2)$ in the sequential limit, where we can find a common ancestor N at which an inequality works for both N_1 and N_2 .

Proposition 25. $f \in \mathcal{R}([a, b], \alpha)$ iff $\forall \varepsilon > 0, \exists P_\varepsilon$ (partition of $[a, b]$) such that $U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) < \varepsilon$.

Proof: ($\Rightarrow$) Assume that $f \in \mathcal{R}([a, b], \alpha)$. Then by properties of *inf* and *sup*, $\forall \varepsilon > 0, \exists P_\varepsilon, P'_\varepsilon$ partition of $[a, b]$ such that

$$\int_a^b f(x) d\alpha - \frac{1}{2}\varepsilon < L(P_\varepsilon, f, \alpha) = \int_a^b f(x) d\alpha \leq \overline{\int_a^b f(x) d\alpha} \leq U(P_\varepsilon, f, \alpha) < \overline{\int_a^b f(x) d\alpha} + \frac{1}{2}\varepsilon \quad (51)$$

Let $P_\varepsilon^* = P_\varepsilon \cup P'_\varepsilon$. Then since $P_\varepsilon^* \supset P_\varepsilon$ and $P_\varepsilon^* \supset P'_\varepsilon$,

$$U(P_\varepsilon^*, f, \alpha) - L(P_\varepsilon^*, f, \alpha) \leq U(P'_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) \quad (52)$$

$$< \left(\int_a^b f(x) d\alpha + \frac{1}{2}\varepsilon \right) - \left(\int_a^b f(x) d\alpha - \frac{1}{2}\varepsilon \right) \quad (53)$$

$$= \int_a^b f(x) d\alpha - \int_a^b f(x) d\alpha + \varepsilon \quad (54)$$

$$= \varepsilon \quad (55)$$

where $\int_a^b f(x) d\alpha - \int_a^b f(x) d\alpha = 0$ since $f \in \mathcal{R}([a, b], \alpha)$.

($\Leftarrow$) Assume that $\forall \varepsilon > 0, \exists P_\varepsilon$ partition of $[a, b]$ such that $U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) < \varepsilon$. We have

$$L(P_\varepsilon, f, \alpha) \leq \int_a^b f(x) d\alpha \leq \int_a^b f(x) d\alpha \leq U(P_\varepsilon, f, \alpha) \quad (56)$$

It follows that

$$0 \leq \int_a^b f(x) d\alpha - \int_a^b f(x) d\alpha \leq U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) \leq \varepsilon \quad (57)$$

Letting $\varepsilon \rightarrow 0$, we have

$$\int_a^b f(x) d\alpha - \int_a^b f(x) d\alpha = 0 \quad (58)$$

therefore $f \in \mathcal{R}([a, b], \alpha)$. □

Thus, the gap on a further refinement (like larger n in sequential proof) is less than the gap on the previous partitions.

Look at Question 2 in Assignment 6.

Theorem 11 (Riemann-Stieltjes integrability). *Let $f, \alpha: [a, b] \rightarrow \mathbb{R}$ be such that f is bounded in $[a, b]$ and α increasing on $[a, b]$.*

Let D be the set of points in $[a, b]$ where f is not continuous. Assume that D is countable and that α is continuous at every point in D . Then $f \in \mathcal{R}([a, b], \alpha)$.

Let $\varepsilon > 0$. For every $x \in [a, b] \setminus D$, since f is continuous at x , there exists $\delta_x > 0$ such that $|f(y) - f(x)| < c\varepsilon$ for all $y \in (x - \delta_x, x + \delta_x) \cap [a, b]$, where the constant $c > 0$ will be determined later.

Since D is countable, $\exists (x_k)_k$ in $[a, b]$ such that $D = \{x_k : k \in \mathbb{N}\}$. Since α is continuous at x_k , there exists $\delta_k > 0$ such that $|\alpha(y) - \alpha(x_k)| < c_k\varepsilon$ for all $y \in (x_k - \delta_k, x_k + \delta_k) \cap [a, b]$, where the constant $c_k > 0$ will be determined later.

Since $[a, b]$ is compact, $[a, b] \subset \bigcup_{x \in [a, b]} (x - \delta_x, x + \delta_x)$, there exists a finite subcover, i.e. $\exists y_1 < \dots < y_{N_\varepsilon} \in [a, b]$ such that

$$[a, b] \subset \bigcup_{i=1}^{N_\varepsilon} (y_i - \delta_{y_i}, y_i + \delta_{y_i}) \quad (59)$$

Continuous functions are always Riemann-Stieltjes integrable. If the function is not continuous, if the set of discontinuity points is countable, and if α is continuous at every discontinuity point, then the function is also Riemann-Stieltjes integrable.

δ_x is defined for every $x \in [a, b]$, since either it's defined by the continuity of f at x (if $x \in [a, b] \setminus D$) or by the continuity of α at x (if $x \in D$).

By ordering the points $a, b, y_i - \delta_{\varepsilon, y}$ (if $\geq a$), and $y_i + \delta_{\varepsilon, y}$ (if $\leq b$), we then obtain a partition $P_\varepsilon = \{x_0, \dots, x_{m_\varepsilon}\}$ such that $\forall j \in \{0, \dots, m_\varepsilon\}$, $\exists i_j \in \{1, \dots, N_\varepsilon\}$ such that $(x_{j-1}, x_j) \subset (y_{i_j} - \delta_{y_{i_j}}, y_{i_j} + \delta_{y_{i_j}}) \cap [a, b]$.

We again separate this into two cases (depending on whether $y_{i_j} \in D$ or not):

- If $y_{i_j} \notin D$, then

$$M_j - m_j = \sup_{x \in [x_{j-1}, x_j]} f(x) - \inf_{x \in [x_{j-1}, x_j]} f(x) \quad (60)$$

$$\leq \left(\sup_{x \in [x_{j-1}, x_j]} f(x) - f(y_{i_j}) \right) + \left(f(y_{i_j}) - \inf_{x \in [x_{j-1}, x_j]} f(x) \right) \quad (61)$$

$$< 2c\varepsilon \quad (62)$$

- If $y_{i_j} \in D$, then we have

$$M_j - m_j \leq M - m = \sup_{x \in [a, b]} f(x) - \inf_{x \in [a, b]} f(x) \quad (63)$$

$$\Delta\alpha_j = \alpha(x_j) - \alpha(x_{j-1}) \leq |\alpha(x_j) - \alpha(y_{i_j})| + |\alpha(y_{i_j}) - \alpha(x_{j-1})| \quad (64)$$

$$< 2c_k\varepsilon \quad (65)$$

Therefore,

$$U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) = \sum_{j=1}^{m_\varepsilon} (M_j - m_j) \Delta\alpha_j \quad (66)$$

$$= \sum_{j=1, y_{i_j} \in D}^{m_\varepsilon} (M_j - m_j) \Delta\alpha_j + \sum_{j=1, y_{i_j} \notin D}^{m_\varepsilon} (M_j - m_j) \Delta\alpha_j \quad (67)$$

$$\leq 2c\varepsilon \underbrace{\sum_{j=1, y_{i_j} \notin D}^{m_\varepsilon} \Delta\alpha_j}_{\leq \sum_{j=1}^{m_\varepsilon} \Delta\alpha_j = \alpha(b) - \alpha(a)} + (M - m) \underbrace{\sum_{j=1, y_{i_j} \in D}^{m_\varepsilon} \Delta\alpha_j}_{\leq 2\varepsilon \sum_{k=1}^{\infty} c_k} \quad (68)$$

the second inequality comes from the fact that $D = \{d_k : k \in \mathbb{N}\}$ and $\Delta\alpha_j < 2c_k\varepsilon$ when $y_{i_j} = d_k$.

Therefore, if we choose $c = \frac{1}{4(\alpha(b) - \alpha(a))}$ and $c_k = \frac{1}{4(M - m)2^k}$ (if $m < M$), so that $2(M - m) \sum_{k=1}^{\infty} c_k = \frac{1}{2}$, then we have

$$U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) < \frac{1}{2}\varepsilon + \frac{1}{2}\varepsilon = \varepsilon \quad (69)$$

Since this is true for all $\varepsilon > 0$, it follows that $f \in \mathcal{R}([a, b], \alpha)$. $\square$ One of the critical part of the proof is to create a shrinking c_k to control error on infinitely many discontinuity points.

Important trick here. And i_j is not a subsequence here, but a function that maps the partition index j to the index of the subcover i_j .

In the discontinuous regime, we have no control on the gap $M_j - m_j$, however, we can use the continuity of α to control the gap $\Delta\alpha_j$.

(Ask in Office Hour)

Proposition 26. Let I be an interval in $\mathbb{R}$ and $f : I \rightarrow \mathbb{R}$ be monotone on I and D be the set of points at which f is not continuous, then D is countable.

Proof: Assuming that f is increasing (otherwise we can consider $-f$). For each $x \in D$, we have that $\lim_{t \rightarrow x^-} f(t)$ and $\lim_{t \rightarrow x^+} f(t)$ exist since f is increasing. Moreover,

$$\lim_{t \rightarrow x^-} f(t) < \lim_{t \rightarrow x^+} f(t) \quad (70)$$

since $x \in D$. By density of $\mathbb{Q}$, it follows that $\exists q_x \in \mathbb{Q}$ such that

$$\lim_{t \rightarrow x^-} f(t) < q_x < \lim_{t \rightarrow x^+} f(t) \quad (71)$$

Moreover, since f is increasing, if $x < y$, then

$$q_x \leq \lim_{t \rightarrow x^+} f(t) \leq \lim_{t \rightarrow y^-} f(t) < q_y \quad (72)$$

Therefore, the map $(D \rightarrow \mathbb{Q}) x \mapsto q_x$ is injective. Since $\mathbb{Q}$ is countable, D is countable. $\square$

Proposition 27 (Integrability of composition). Let $a, b, c, d \in \mathbb{R}$ be such that $a < b, c < d$, $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing, and $f : [a, b] \rightarrow [c, d]$ and $\Phi : [c, d] \rightarrow \mathbb{R}$. If $f \in \mathcal{R}([a, b], \alpha)$ and Φ is continuous, then $\Phi \circ f \in \mathcal{R}([a, b], \alpha)$.

Proof: Since $[c, d]$ is compact and Φ continuous on $[c, d]$, we obtain that Φ is uniformly continuous on $[c, d]$. Thus, $\forall \varepsilon > 0$, $\exists \delta_\varepsilon > 0$ such that $\forall x, y \in [c, d]$,

$$|x - y| < \delta_\varepsilon \implies |\Phi(x) - \Phi(y)| < \lambda \varepsilon \quad (73)$$

where the constant $\lambda > 0$ will be determined later.

On the other hand, since $f \in \mathcal{R}([a, b], \alpha)$, $\exists P_\varepsilon$ partition of $[a, b]$ such that

$$U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) = \sum_{i=1}^n (M_i - m_i) \Delta \alpha_i < \lambda \varepsilon \quad (74)$$

where the constant $\lambda_\varepsilon > 0$ will be determined later. We want to show that

$$U(P_\varepsilon, \Phi \circ f, \alpha) - L(P_\varepsilon, \Phi \circ f, \alpha) = \sum_{i=1}^n (M'_i - m'_i) \Delta \alpha_i < \varepsilon \quad (75)$$

and we need to estimate the term product $(M'_i - m'_i) \Delta \alpha_i$ for each $i \in \{1, \dots, n\}$.

For all $i \in \{1, \dots, n\}$, we have

$$M'_i - m'_i \leq \begin{cases} \lambda \varepsilon & \text{if } M_i - m_i < \delta_\varepsilon \\ M - m & \text{if } M_i - m_i \geq \delta_\varepsilon \end{cases} \quad (76)$$

This shows that the countable discontinuity condition in the previous theorem is automatically satisfied for monotone functions.

Remark 7. If α is continuous, then every monotone function is Riemann-Stieltjes integrable with respect to α .

Implication: if we have a function f that is Riemann-Stieltjes integrable, is the absolute value, or power of f also Riemann-Stieltjes integrable?

The trick is to keep a constant λ like the proof before.

Recall that $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$ and $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$.

where $M = \sup_{x \in [c,d]} \Phi(x)$ and $m = \inf_{x \in [c,d]} \Phi(x)$. This gives:

$$\sum_{i=1}^n (M'_i - m'_i) \Delta \alpha_i \leq \underbrace{\lambda \varepsilon \sum_{i=1, M_i - m_i < \delta_\varepsilon}^n \Delta \alpha_i}_{\leq \lambda \varepsilon (\alpha(b) - \alpha(a))} + (M - m) \sum_{i=1, M_i - m_i \geq \delta_\varepsilon}^n \Delta \alpha_i \quad (77)$$

The first term is easy to control, the second term is trickier. But we haven't use the Riemann-Stieltjes integrability of f yet. We have:

$$(M - m) \sum_{i=1, M_i - m_i \geq \delta_\varepsilon}^n \Delta \alpha_i \leq (M - m) \sum_{i=1, M_i - m_i \geq \delta_\varepsilon}^n \frac{M_i - m_i}{\delta_\varepsilon} \Delta \alpha_i \quad (78)$$

Since $M_i - m_i \geq \delta_\varepsilon$. Since $\sum_{i=1}^n (M_i - m_i) \Delta \alpha_i < \lambda_\varepsilon$, we have

$$(M - m) \sum_{i=1, M_i - m_i \geq \delta_\varepsilon}^n \Delta \alpha_i \leq \frac{M - m}{\delta_\varepsilon} \sum_{i=1}^n (M_i - m_i) \Delta \alpha_i < \frac{M - m}{\delta_\varepsilon} \lambda_\varepsilon \quad (79)$$

Now, choosing $\lambda = \frac{1}{2(\alpha(b) - \alpha(a))}$ and $\lambda_\varepsilon = \frac{\delta_\varepsilon \varepsilon}{2(M - m)}$, we have

$$\sum_{i=1}^n (M'_i - m'_i) \Delta \alpha_i \leq \frac{1}{2} \varepsilon + \frac{1}{2} \varepsilon = \varepsilon \quad (80)$$

In the first case, as soon as we have one ε in the product term, it will be small.

In the second case, we can only say that $M'_i - m'_i$ is bounded, but fortunately, we have control on the other term $\Delta \alpha_i$.

We introduce the term $\frac{M_i - m_i}{\delta_\varepsilon}$ so that we can use the fact that $\sum_{i=1}^n (M_i - m_i) \Delta \alpha_i < \lambda_\varepsilon$ from the integrability.

□

Proposition 28. Let $a, b \in \mathbb{R}$ be such that $a < b$ and $\alpha : [a, b] \rightarrow \mathbb{R}$.

1. $\forall f, g \in \mathcal{R}([a, b], \alpha)$, $f + g \in \mathcal{R}([a, b], \alpha)$
and

$$\int_a^b (f + g) d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha \quad (81)$$

2. $\forall \lambda \in \mathbb{R}, f \in \mathcal{R}([a, b], \alpha)$, we have $\lambda f \in \mathcal{R}([a, b], \alpha)$ and

$$\int_a^b \lambda f d\alpha = \lambda \int_a^b f d\alpha \quad (82)$$

3. $\forall f, g \in \mathcal{R}([a, b], \alpha)$ such that $f(x) \leq g(x)$ on $[a, b]$ implies that

$$\int_a^b f d\alpha \leq \int_a^b g d\alpha \quad (83)$$

4. $\forall f \in \mathcal{R}([a, b], \alpha)$, $|f| \in \mathcal{R}([a, b], \alpha)$ and

$$\left| \int_a^b f d\alpha \right| \leq \int_a^b |f| d\alpha \quad (84)$$

5. $\forall f, g \in \mathcal{R}([a, b], \alpha)$, $fg \in \mathcal{R}([a, b], \alpha)$ (but we don't have a simple formula for $\int_a^b fg d\alpha$).

6. $\forall f \in \mathcal{R}([a, b], \alpha)$, $c \in (a, b)$ we have

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha \quad (85)$$

Proof: (1.) $\forall P, P'$ partition of $[a, b]$, letting $P^* = P \cup P'$, we have

$$\overline{\int_a^b (f + g) d\alpha} \leq U(P^*, f + g, \alpha) \quad (86)$$

since $\sup(f + g) \leq \sup f + \sup g$

$$\leq U(P^*, f, \alpha) + U(P^*, g, \alpha) \quad (87)$$

By taking the infimum over P and P' , we have

$$\overline{\int_a^b (f + g) d\alpha} \leq \overline{\int_a^b f d\alpha} + \overline{\int_a^b g d\alpha} \quad (88)$$

$$= \int_a^b f d\alpha + \int_a^b g d\alpha \quad (f, g \in \mathcal{R}([a, b], \alpha)) \quad (89)$$

Similarly, we can show that

$$\underline{\int_a^b (f + g) d\alpha} \geq \underline{\int_a^b f d\alpha} + \underline{\int_a^b g d\alpha} \quad (90)$$

Therefore,

$$\int_a^b f d\alpha + \int_a^b g d\alpha \leq \int_a^b (f+g) d\alpha \leq \overline{\int_a^b (f+g) d\alpha} \leq \int_a^b f d\alpha + \int_a^b g d\alpha \quad (91)$$

hence $f+g \in \mathcal{R}([a, b], \alpha)$ and

$$\int_a^b (f+g) d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha \quad (92)$$

(2.) follows from

$$\sup(\lambda f) = \lambda \sup f \text{ and } \inf(\lambda f) = \lambda \inf f \text{ if } \lambda \geq 0 \quad (93)$$

and $\sup(-f) = -\inf(f)$, $\inf(-f) = -\sup(f)$. This gives

$$U(P, -f, \alpha) = -L(P, f, \alpha) \text{ and } L(P, -f, \alpha) = -U(P, f, \alpha) \quad (94)$$

$$\overline{\int_a^b (-f) d\alpha} = -\int_a^b f d\alpha \text{ and } \underline{\int_a^b (-f) d\alpha} = -\overline{\int_a^b f d\alpha} \quad (95)$$

$$(3.) \text{ follows from } f \leq g \implies \begin{cases} \sup f \leq \sup g \\ \inf f \leq \inf g \end{cases}.$$

(4.) follows from absolute function is continuous, so $|f|$ is Riemann-Stieltjes integrable. We apply (2. & 3.) with $-|f| \leq f \leq |f|$ to obtain the desired inequality.

(5.) Obverse that $fg = \frac{1}{2}((f+g)^2 - f^2 - g^2)$. Using (1, 2) and the fact that $x \mapsto x^2$ is continuous.

(6.) Assignment 6. □

We have to use two side inequality to bound absolute value.

Proposition 29. Let $a, b \in \mathbb{R}$ such that $a < b$ and $\alpha, \alpha_1, \alpha_2 : [a, b] \rightarrow \mathbb{R}$ be increasing on $[a, b]$. Then:

1. $\forall f \in \mathcal{R}([a, b], \alpha_1) \cap \mathcal{R}([a, b], \alpha_2)$, $f \in \mathcal{R}([a, b], \alpha_1 + \alpha_2)$ and

$$\int_a^b f d(\alpha_1 + \alpha_2) = \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2 \quad (96)$$

2. $\forall f \in \mathcal{R}([a, b], \alpha)$, $\lambda \in (0, \infty)$, $f \in \mathcal{R}([a, b], \lambda\alpha)$ and

$$\int_a^b f d(\lambda\alpha) = \lambda \int_a^b f d\alpha \quad (97)$$

Theorem 12. Let $a, b \in \mathbb{R}$ be such that $a < b$ and $f \in \mathcal{R}([a, b])$. Then the function $F : [a, b] \rightarrow \mathbb{R}$ defined by $F(x) = \int_a^x f$, $\forall x \in [a, b]$ is lipchitz continuous on $[a, b]$. Moreover, if f is continuous at $x_0 \in [a, b]$, then F is differentiable at x_0 and

$$F'(x_0) = f(x_0) \quad (98)$$

Proof: $\forall x, y \in [a, b]$,

$$|F(x) - F(y)| \leq \left| \int_a^y f - \int_a^x f \right| = \left| \int_x^y f \right| \quad (99)$$

$$\leq \int_x^y |f| \leq \left(\sup_{t \in [a, b]} |f(t)| \right) |y - x| \quad (100)$$

Thus F is Lipschitz continuous on $[a, b]$.

If f is continuous at x_0 , then for all $\varepsilon > 0$, $\exists \delta_\varepsilon > 0$ such that for all $x \in [a, b]$, we have

$$|f(x) - f(x_0)| < \varepsilon \text{ whenever } |x - x_0| < \delta_\varepsilon \quad (101)$$

hence for every $t \in (-\delta_\varepsilon, \delta_\varepsilon) \setminus \{0\}$ such that $x_0 + t \in [a, b]$, we have

$$\left| \frac{1}{t} (F(x_0 + t) - F(x_0)) - f(x_0) \right| = \left| \frac{1}{t} \left(\int_{x_0}^{x_0+t} f \right) - \frac{1}{t} \int_{x_0}^{x_0+t} f(x_0) \right| \quad (102)$$

$$= \left| \frac{1}{t} \int_{x_0}^{x_0+t} (f(s) - f(x_0)) ds \right| \quad (103)$$

$$\leq \frac{1}{|t|} \int_{x_0}^{x_0+t} |f(s) - f(x_0)| ds \quad (104)$$

$$\leq \frac{1}{|t|} \int_{x_0}^{x_0+t} \varepsilon = \varepsilon \quad (105)$$

where $\int_{x_0}^{x_0+t} = -\int_{x_0+t}^{x_0}$ if $t < 0$.

It follows that,

$$\limsup_{t \rightarrow 0} \left| \frac{1}{t} (F(x_0 + t) - F(x_0)) - f(x_0) \right| \leq \varepsilon \quad (106)$$

Letting $\varepsilon \rightarrow 0$ (which is allowed since limsup does not depends on t), we have

$$\lim_{t \rightarrow 0} \left| \frac{1}{t} (F(x_0 + t) - F(x_0)) - f(x_0) \right| = 0 \quad (107)$$

i.e, F is differentiable at x_0 and $F'(x_0) = f(x_0)$. $\square$

Integrability guarantees Lipschitz continuity and continuity guarantees differentiability.

This is how you bound things at a limit point.

Theorem 13 (Fundamental theorem of calculus). *Let $a, b \in \mathbb{R}$ be such that $a < b$ and $F : [a, b] \rightarrow \mathbb{R}$ be such that F is continuous on $[a, b]$, differentiable on (a, b) , and $F' = f \in \mathcal{R}([a, b])$. Then*

$$\int_a^b f(x) = F(b) - F(a) \quad (108)$$

Theorem 14 (Taylor's theorem with integral reminder). *Let $x, x_0 \in \mathbb{R}$, $x \neq x_0$, $I = \begin{cases} [x_0, x] & \text{if } x > x_0 \\ [x, x_0] & \text{if } x < x_0 \end{cases}$, $\text{int}(I) = \begin{cases} (x_0, x) & \text{if } x > x_0 \\ (x, x_0) & \text{if } x < x_0 \end{cases}$ and $f : I \rightarrow \mathbb{R}$ be n times continuously differentiable in I (i.e. $f^{(n)}$ exists and is continuous in I), $n + 1$ times differentiable in $\text{int}(I)$, and $t \mapsto f^{(n+1)}(t)(x - t)^n \in \mathcal{R}(I)$.*

Then,

$$f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k}_{\text{partial sum of Taylor series at } x_0} + \underbrace{\int_{x_0}^x \frac{f^{(n+1)}(t)}{n!} (x - t)^n dt}_{\text{integral reminder}} \quad (109)$$

Proof In the finite version of Taylor's theorem, we introduce $h(t) = g(t)(x - x_0)^{n+1} - g(x_0)(x - t)^{n+1}$ with

$$g(t) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (t - t)^k \quad (110)$$

and we proved that $h(x) = h(x_0) = 0$ and

$$h'(t) = (n + 1)g(x_0)(x - t)^n - \frac{1}{n!} f^{(n+1)}(t)(x - t)^n (x - x_0)^{n+1} \quad (111)$$

From the assumption on f , we have that h is continuous on I , differentiable on $\text{int}(I)$, and $h'(t) \in \mathcal{R}(I)$. By applying the Fundamental Theorem of Calculus to h' , we then obtain

$$0 = h(x) - h(x_0) = \int_{x_0}^x h'(t) dt = \underbrace{g(x_0)(x - x_0)^{n+1}}_{=g(x_0)[-(x-t)^{n+1}]_{x_0}^x} - \int_{x_0}^x \frac{f^{(n+1)}(t)}{n!} (x - t)^n (x - x_0)^{n+1} dt \quad (112)$$

$$= (x - x_0)^{n+1} \left(g(x_0) - \int_{x_0}^x \frac{f^{(n+1)}(t)}{n!} (x - t)^n dt \right) \quad (113)$$

Hence

$$g(x_0) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k = \int_{x_0}^x \frac{f^{(n+1)}(t)}{n!} (x - t)^n dt \quad (114)$$

□

If $f^{(n+1)}(t)(x - t)^n$ is not defined at the endpoints, we assume it equals some Riemann integrable function at the endpoints.

Definition 19 (Riemann-Stieltjes integrable). Let $a \in \mathbb{R}$, $b \in (a, \infty)$, or $b = \infty$, and $\alpha : [a, b) \rightarrow \mathbb{R}$ be increasing. We say that $f : [a, b) \rightarrow \mathbb{R}$ is Riemann-Stieltjes integrable with respect to α on $[a, b)$ if $f \in \mathcal{R}([a, s], \alpha)$ for all $s \in (a, b)$ and the limit

$$\lim_{s \rightarrow b^-} \int_a^s f d\alpha \quad (115)$$

exists and is finite. We denote

$$\int_a^b f d\alpha \left(\text{or } \int_a^{b^-} f d\alpha \text{ if } b < \infty \right) = \lim_{s \rightarrow b^-} \int_a^s f d\alpha \quad (116)$$

which we call R-S integral of f over $[a, b)$ with respect to α .

Similarly, for $b \in \mathbb{R}$, $a \in (-\infty, b)$ or $a = -\infty$, and $\alpha : (a, b] \rightarrow \mathbb{R}$ increasing, we say that $f : (a, b] \rightarrow \mathbb{R}$ is Riemann-Stieltjes integrable with respect to α on $(a, b]$ if $f \in \mathcal{R}([s, b], \alpha)$ for all $s \in (a, b)$ and the limit

$$\lim_{s \rightarrow a^+} \int_s^b f d\alpha \quad (117)$$

exists and is finite. We denote

$$\int_a^b f d\alpha \left(\text{or } \int_{a^+}^b f d\alpha \text{ if } a > -\infty \right) = \lim_{s \rightarrow a^+} \int_s^b f d\alpha \quad (118)$$

which we call R-S integral of f over $(a, b]$ with respect to α .

Proposition 30. Let $a, b \in \mathbb{R}$ be such that $a < b$, $\alpha : [a, b) \rightarrow \mathbb{R}$ be increasing, and $f \in \mathcal{R}([a, b), \alpha)$.

1. If α is continuous at b then

$$\int_a^{b^-} f d\alpha = \int_a^b f d\alpha \quad (119)$$

2. If α is continuous at a , then

$$\int_{a^+}^b f d\alpha = \int_a^b f d\alpha \quad (120)$$

Proof: (1.) $\forall (a, b) \in (a, b)$, we have

$$\left| \int_a^b f d\alpha - \int_a^s f d\alpha \right| = \left| \int_s^b f d\alpha \right| \leq \sup_{x \in [s, b]} |f(x)| \int_s^b 1 d\alpha \quad (122)$$

$$= \sup_{x \in [s, b]} |f(x)| (\alpha(b) - \alpha(s)) \quad (123)$$

Since α is continuous at b , we obtain $\lim_{s \rightarrow b^-} \alpha(s) = \alpha(b)$, hence

$$\lim_{s \rightarrow b^-} \int_a^s f d\alpha = \int_a^b f d\alpha \quad (124)$$

Remark 8. If α is not continuous at a or b , then this is not true in general. For example, if $\alpha = 0$ on $[a, b)$ and $\alpha(b) = 1$, then

$$\int_a^b f d\alpha = 1 \text{ but } \int_a^s f d\alpha = 0 \text{ for all } s \in (a, b) \quad (121)$$

(2.) is similar to (1.). $\square$

This allows us to define integral on functions that are unbounded at the endpoints, e.g. power functions.

Example 5

$$\int_1^s \frac{1}{x^p} dx = \begin{cases} \ln s - \ln 1 = \ln s \rightarrow \infty & \text{if } p = 1 \\ \frac{1}{p-1} (1 - \frac{1}{s^{p-1}}) \rightarrow \begin{cases} \frac{1}{p-1} & \text{if } p > 1 \\ \infty & \text{if } p < 1 \end{cases} & \end{cases} \quad (125)$$

$$\text{Hence, we have } \begin{cases} \int_1^\infty \frac{1}{x^p} dx = \infty & \text{if } p \leq 1 \\ x \mapsto \frac{1}{x^p} \text{ is not Riemann integrable over } [1, \infty) & \text{if } p > 1 \end{cases}.$$

In the example we use $\frac{d}{dx} \ln x = \frac{1}{x}$ which we proved for $0 < x < 2$ where $\ln(x)$ is a power series. Proof when $0 < x < \infty$: Observe that for all $x, y \in (0, \infty)$, we have $e^{\ln x + \ln y} = e^{\ln x} e^{\ln y} = xy$. Hence, we have $\ln x + \ln y = \ln(xy)$ and $\ln x + \ln \frac{1}{x} = \ln 1 = 0$ so $\ln \frac{1}{x} = -\ln x$. It follows that, for every $x \in (0, \infty)$ and t sufficiently small ($|\frac{t}{x}| < 1$),

$$\frac{1}{t} (\ln(x+t) - \ln x) = \frac{1}{t} \ln \left(1 + \frac{t}{x} \right) \quad (126)$$

$$= \frac{1}{t} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \left(\frac{t}{x} \right)^k \quad (127)$$

$$= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \left(\frac{t}{x} \right)^{k-1} \quad (128)$$

which is continuous and converges to $\frac{1}{x}$ at $t = 0$. Hence, $\ln x$ is differentiable at x and $\frac{d}{dx} \ln x = \frac{1}{x}$.

References